COMP3311

Week 2 Tutorial

Agenda

- Admin Stuff
- Week 1 key concepts
- Tutorial questions
- Demo: how to ssh into vxdb02

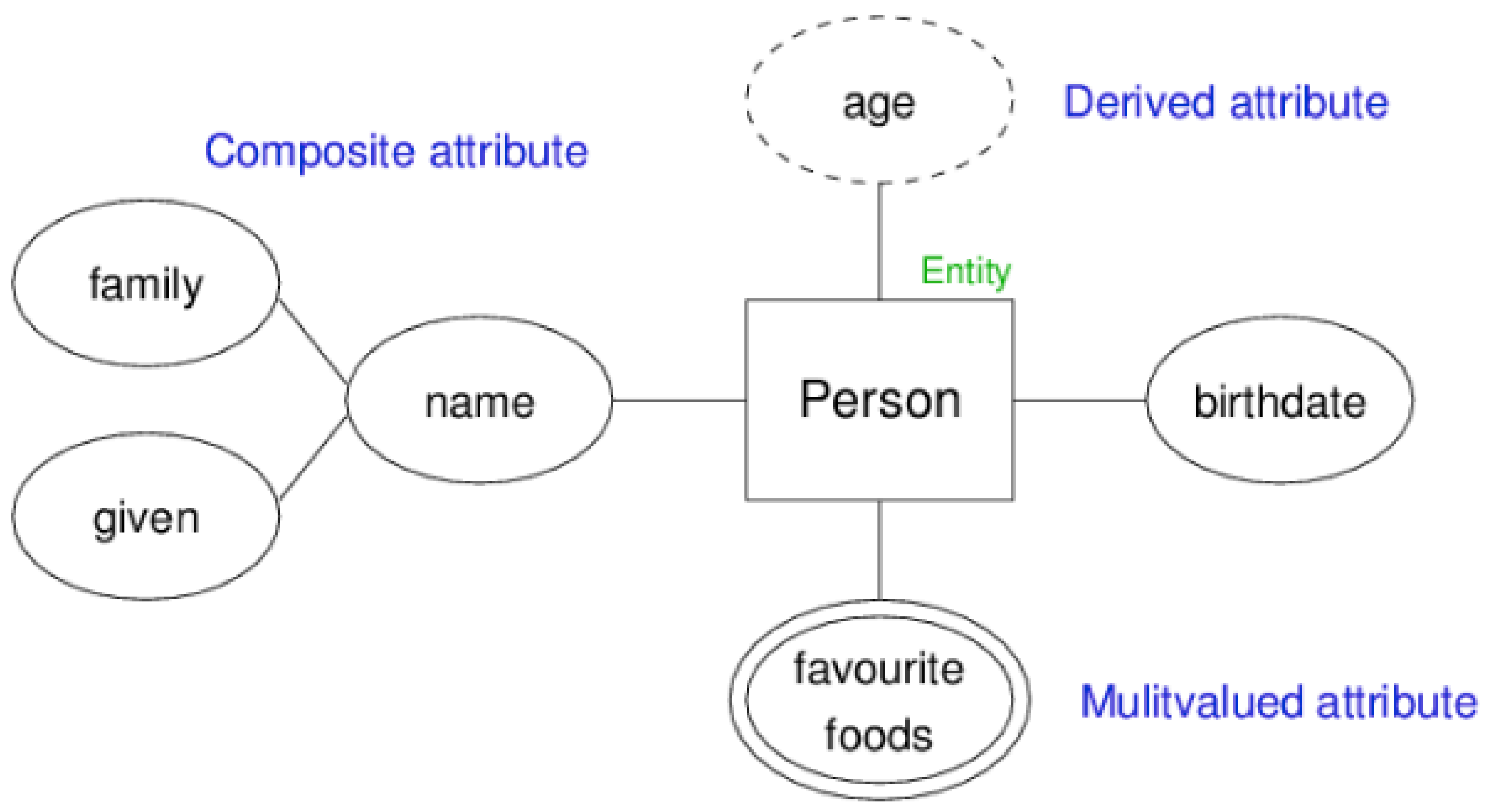
Admin Stuff

- **Getting help:**
  - Discourse (forum)
  - Help sessions (for assignments)
  - Class email (cs3311@cse.unsw.edu.au)
  - Email me (z5479432@ad.unsw.edu.au)
- **Assessment Structure**
  - Quizzes (Weeks 2,3,4,7,8,10) - 12%
  - Assignment 1 (Week 5) - 13%
  - Assignment 2 (Week 10) - 15%
  - Final (Exam period) - 60%
- **Tutorial**
  - 1.5 hr class going through tutorial questions
  - Tutorial questions are based on content from the week before
    - I like to go through the core concepts you'll need to use to answer the questions so please feel free to come along even if you're not caught up to lectures :D
  - Tutorial questions help you prepare for assignments and especially the final exam
  - Solutions for tutorial questions come out the week after
  - To access the work done in these tutorials: https://github.com/nicoleewang/tutoring/tree/main/3311/26t2/H09A

Week 1 Key Concepts

Data Modelling - ER Diagram

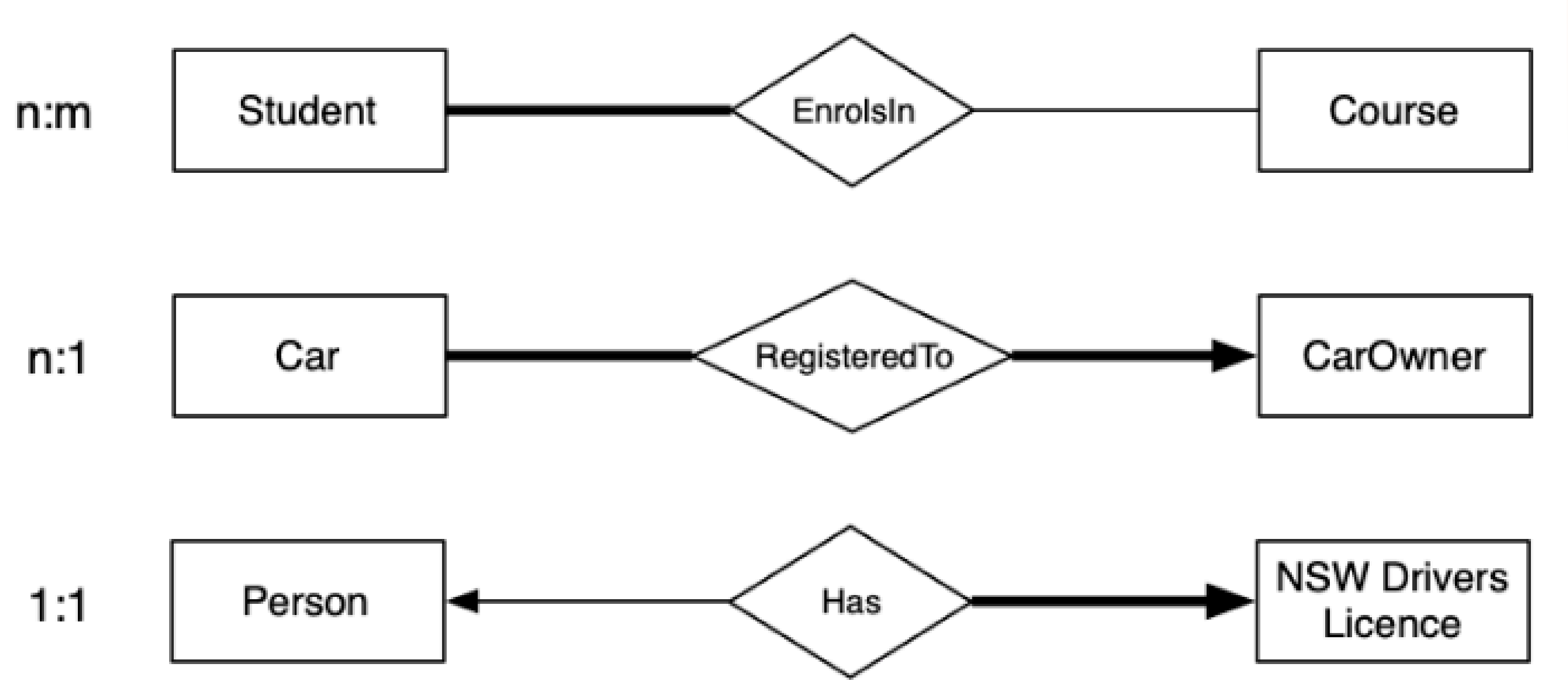
- ER diagram: modelling notation containing collections of
  - entities (**rectangles**) and their associated attributes (**ovals**)
  - relationships (**diamonds**) between entities
- Special types of attributes:



Week 1 Key Concepts

Data Modelling - ER Diagram

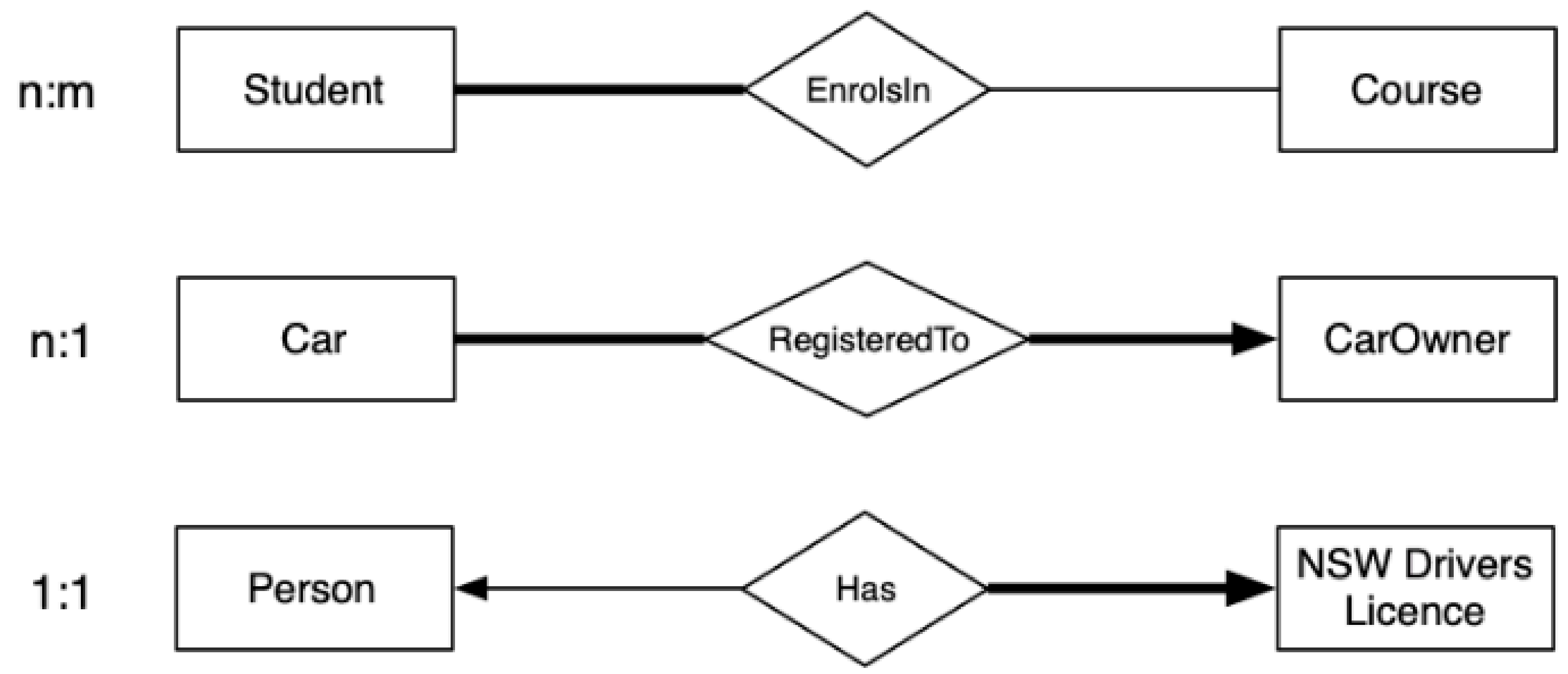
- Relationships
  - Thick/Thin lines represent **participation**
  - Thick line = total participation
    - e.g. (diagram 1) to be a Student you NEED to be enrolled in a course
  - Thin line = partial participation
    - e.g. (diagram 1) a Course can have Students enrolled in it (but not must)
  - How to read the diagram
    - The line next to the entity describes the participation of that entity



Week 1 Key Concepts

Data Modelling - ER Diagram

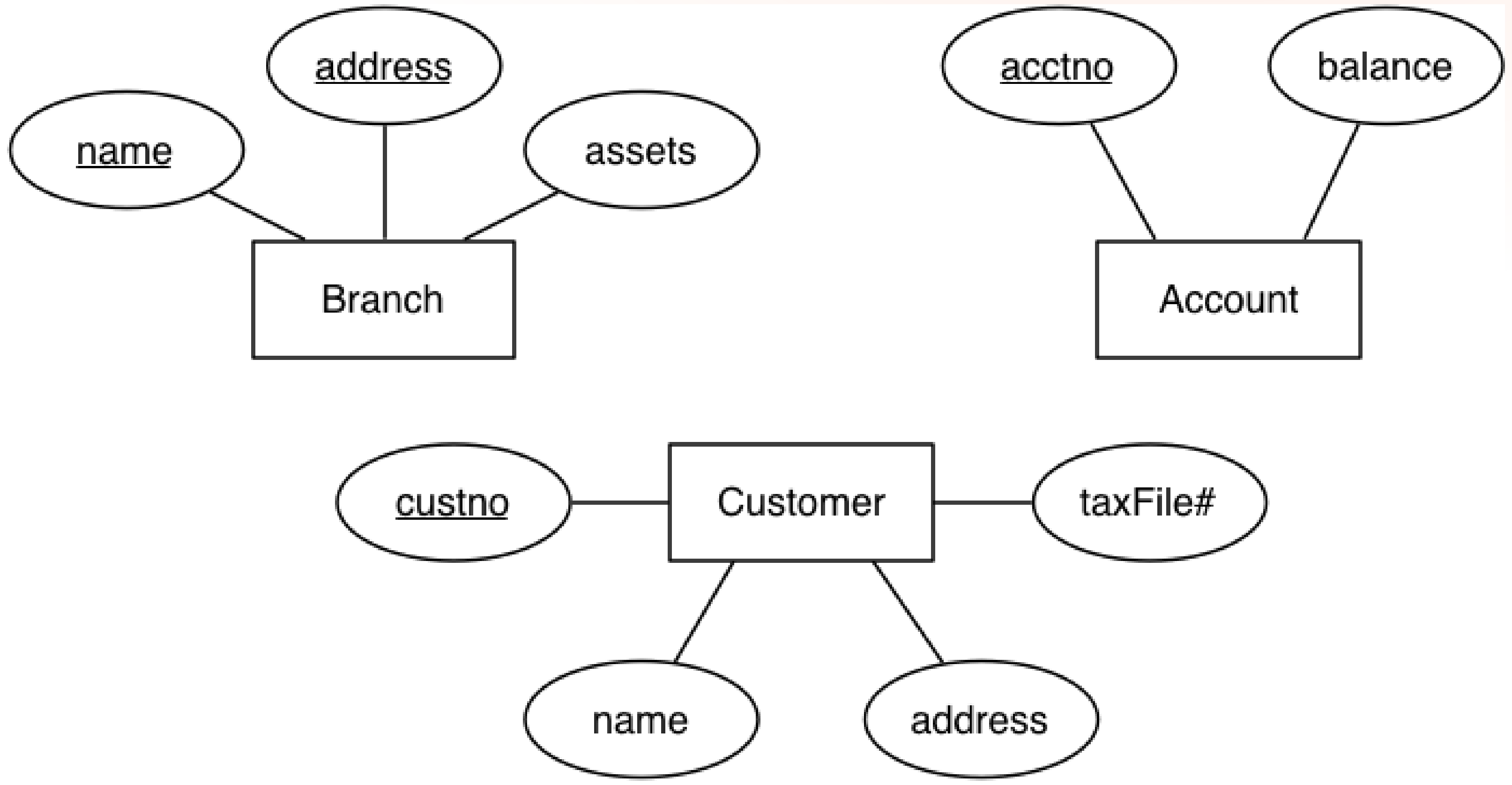
- Relationships
  - Arrows indicate **cardinality**
  - No arrow = any number
    - e.g. (diagram 1) a Student can enrol in as many Courses as they want and a Course can have as many students enrolled in it
  - Arrow = at most 1
    - e.g. (diagram 3) a Person can have at most one NSW Driver Licence
  - How to read the diagram:
    - An arrow touching an entity means that entity participates at most once in the relationship



Week 1 Key Concepts

Data Modelling - ER Diagram

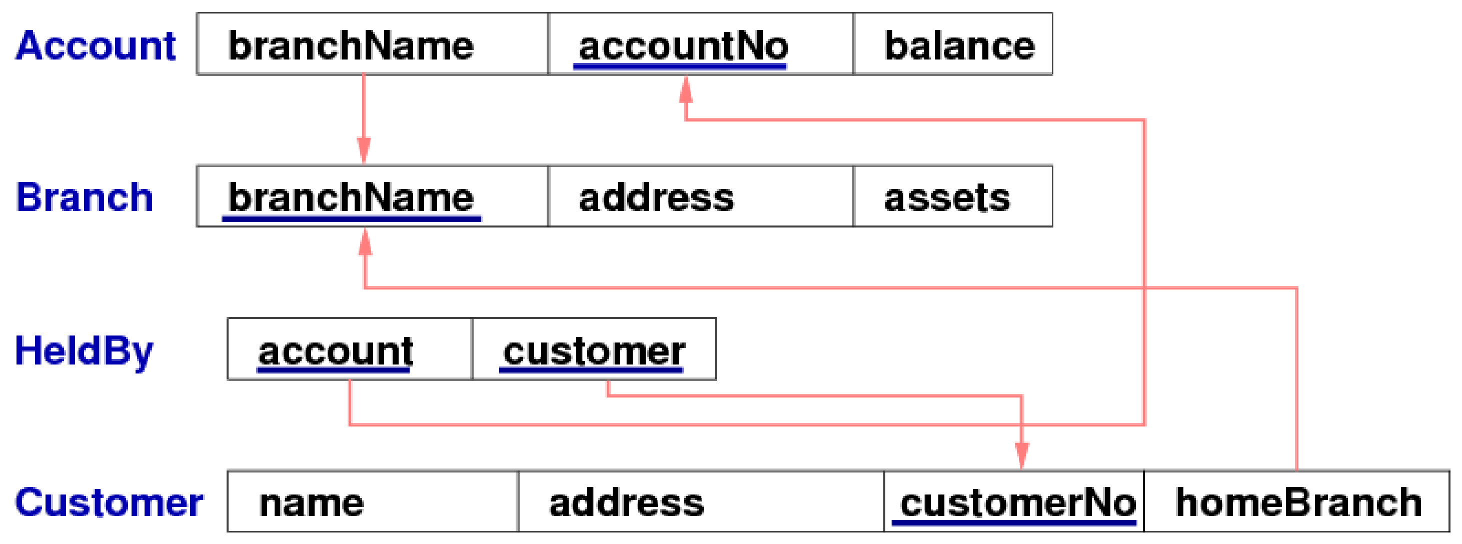
- Primary Keys
    - Entities have attribute(s) that are used to identify entity instances from one another
    - primary keys are NOT NULL and UNIQUE for all instances in the set
    - in an ER diagram: primary keys are indicated by an <u>underline</u>
    - keys with more than one attribute are called composite keys



Week 1 Key Concepts

Data Modelling - Relational Model

- What is it?
  - Relational modelling is a different type of modelling compared to ER diagrams
  - models data as a collection of relations (tables/set of tuples) with attributes
  - Better represents the structure of the schema when converting to SQL
- What does it look like?
  - Relation Schemas (usually denoted R,S,T,...) represent the structure of the relation, and has:
    - A name (unique within the database)
    - A set of attributes (can be seen as column headings for a table)
    - `e.g. Account(branchName, accountNo, balance)`



Tutorial Questions

Now let's try some questions :D

- Q1
- Q7
- Q9
- Q11
- Q15
- Q18

Demo: how to ssh into vxdb02

Setting Up

- Follow prac exercise 01 for detailed instructions on how to set up!
- How to set up PostgreSQL:
  - install on your home computer
  - set it up on your CSE account and access through VLab or SSH (I reccomend this)
- Setup on your CSE account:
  - from cse servers, ssh via:
    - `ssh vxdb02`
  - from home, ssh via:
    - `ssh YourZID@vxdb02.cse.unsw.edu.au`
  - once you ssh run this to start the server (need to do this every time)
    - `source /localstorage/$USER/env`
    - `p1`
- Ending your session
  - please ensure you run the following when you are done
    - `p0`

Demo: how to ssh into vxdb02

Useful Commands

- Within the PostgresSQL server (in a linux shell):
  - `psql -l`
    - to see what databases you have
  - `createdb <db name>`
    - create a new db
  - `psql <db name>`
    - access the database
  - `psql <db name> -f <file name>`
    - to populate your database with a SQL file
  - `p1`
    - start server
  - `p0`
    - stop server
- Inside psql (within a database):
  - `\d`
    - to find relations within your database
  - `\d <table name>`
    - to find more details about an individual table
  - `\q`
    - quit
  - other SQL queries…

Reminders for the Week

Have you done these things???

- Quiz 1 (Due this Friday 11:59pm)
- Try to ssh into vxdb02 on your machine, run the server and play around with it